

AIOS

v2

A multi-tenant SEO operations platform, rebuilt from zero to be measurably honest, durable under failure, and impossible to build carelessly.

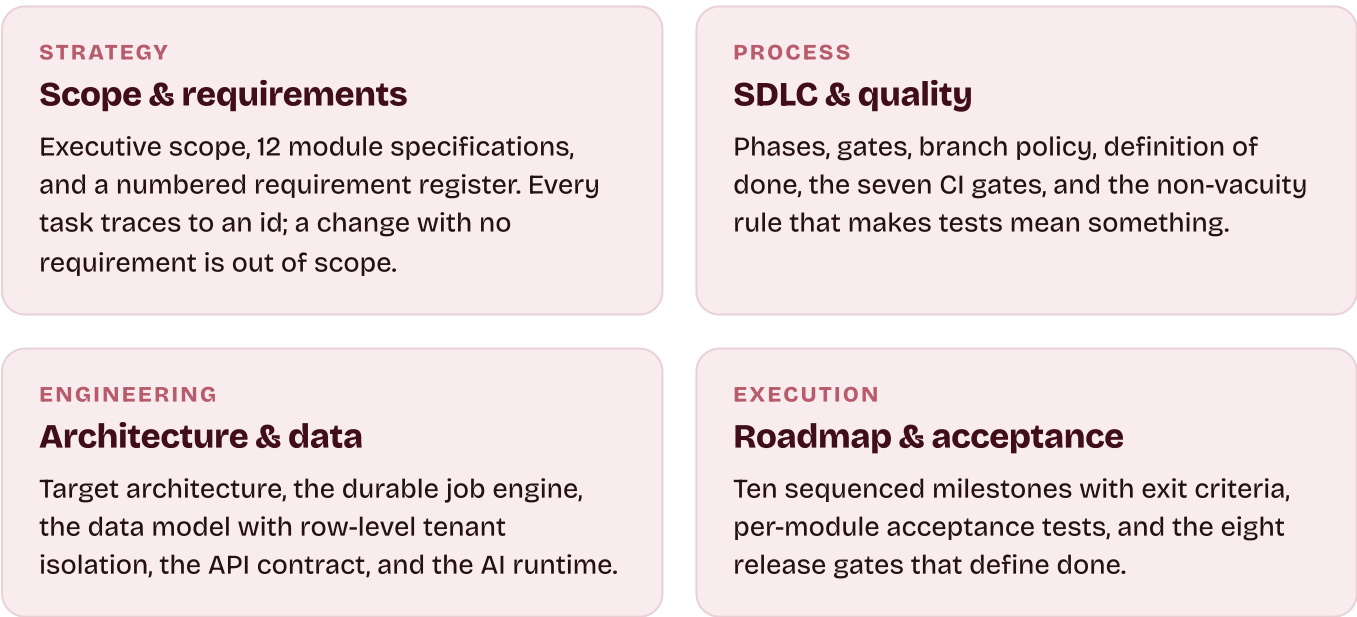
THE BRIEF

One platform. Twelve modules. Fifty to a hundred clients.

An agency runs its entire SEO delivery here: auditing, keyword research, content written in the client's own design language and published as editable pages, citation building, off-page placement across 32 platforms, tracking, and reporting.



What this document set contains



This is a rebuild, not a refactor. v1 exists, runs, and proved the domain. Its defects are structural: data that was synthesised and shown as measured, jobs that reported success having done nothing, and design capture that could not produce an editable page. v2 keeps the domain knowledge and discards the architecture.

Twelve modules, one spine

All mount on a shared core: one identity, one job engine, one cost gate, one vault, one provenance layer.

M01 Portal & Core Identity, RBAC, clients, NAP, tasks, reviews, vault, cost control, the job engine.	M02 Audit Free lead-magnet audit plus six paid types. One findings source, two renderers.	M03 Content System Learn the design language, plan from a topical map, write, QA, publish as editable Elementor.
M04 Citations Extension + AI agent that creates accounts and fills forms. A human presses submit.	M05 Web 2.0 & Social 32 platforms, official APIs, per-client identity, anchors, pacing, link liveness.	M06 Policy Radar Watch the industry, detect changes, map exposure per client, brief daily.
M07 Rank & Grid Organic rank and local geo-grid. Sources labelled, never mixed. Three-state points.	M08 Keyword Research The single upstream of the word bank: volumes, intent, clusters, topical map.	M09 Indexing Submit to Google, IndexNow and Bing with quota queueing. Verify indexation.
M10 Local & GBP Profile, posts, reviews, Q&A, and the canonical NAP citations propagate.	M11 Reporting One document model. HTML and PDF are two renderers, never two templates.	M12 Billing & Cost Tiers, entitlements, money dials, caps, and the loaded cost of every unit.

Ten rules. Each one is a defect v1 shipped.

These are not style preferences. A change that violates one is rejected in review regardless of whether its tests pass.

P1**Never invent data**

An unconfigured provider returns a **degraded result** — never a plausible number. Enforced by a CI gate over every writing path.

P2**Three states, never two**

Measured, **absent** and **unmeasured** are different facts. Every ratio divides by measured only.

P3**Degrade, never crash**

Every external seam returns a typed result with a machine-branchable reason. No caller parses a message string.

P4**Terminal states are honest**

A publish with no credentials is **blocked**, not done. Success requires the effect to be **confirmed**.

P5**Money is gated first**

The estimate gates before the call; the actual is committed after. A blocked call never reaches the provider.

P6**The database enforces tenancy**

Forced row-level security on every client table. Application code is the second line of defence, never the first.

P7**Jobs are idempotent**

Durable, resumable, retried with bounds, dead-lettered with context. Re-running never double-spends or double-publishes.

P8**Provenance on every value**

Measured, derived, declared, inferred or defaulted — with source and timestamp. No provenance, no display.

P9**Secrets are sealed**

Envelope-encrypted, per client, audited on read. Never in a log, a response, a job payload or a trace.

P10**Tests prove behaviour**

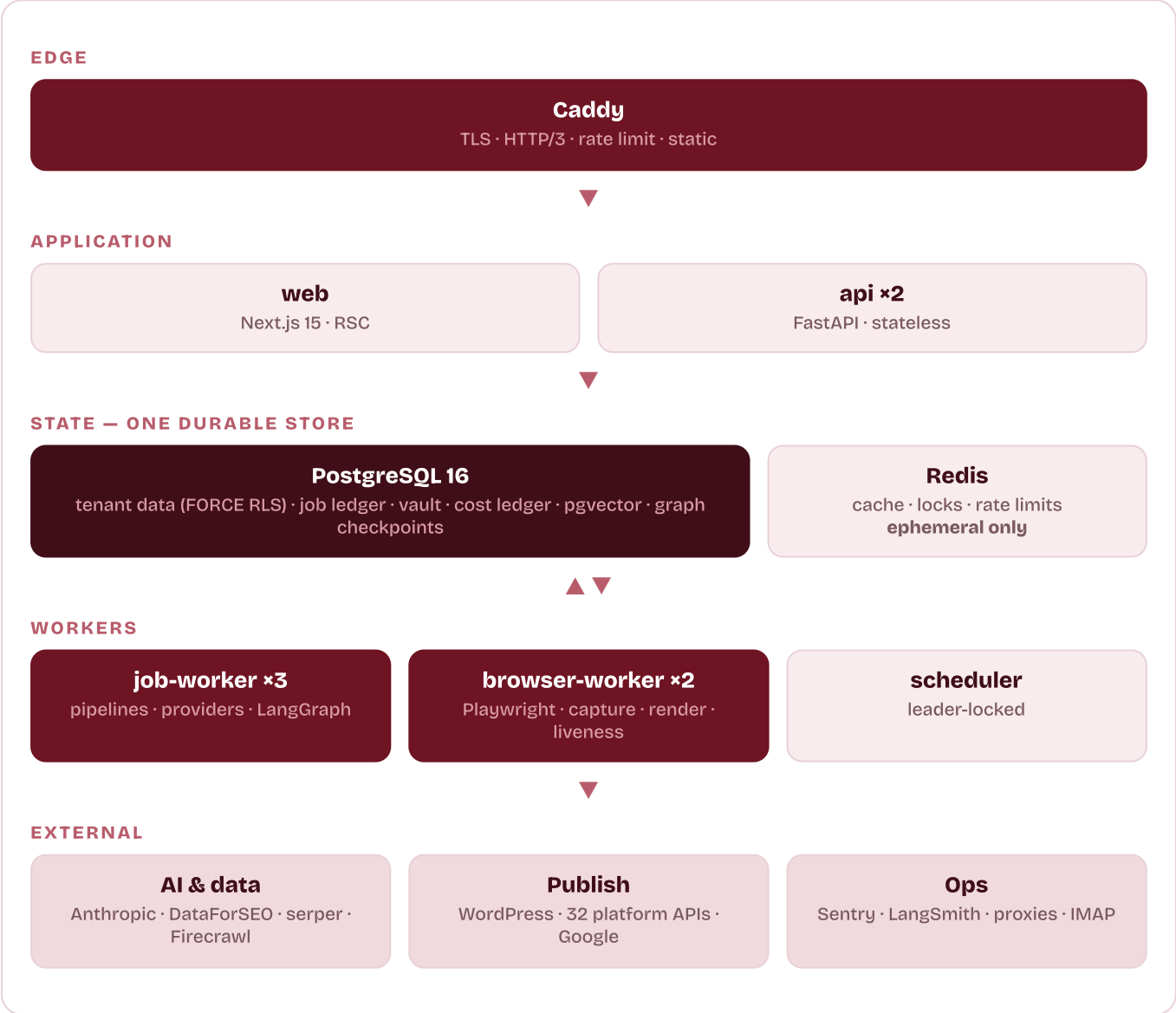
Every test must **fail when its defect is re-injected**. One that passes against a broken implementation is deleted, not kept.

v1 had roughly 4,900 tests and still shipped a form-filler that reported success while filling a honeypot.

The test could not fail: the environment made its branch unreachable. Test count is not the metric — non-vacuity is.

A modular monolith with detachable workers

One VPS. Modules are enforced package boundaries, not services — the isolation without the distributed-system tax.



One store

A job's state and its side-effect rows commit in the **same transaction**. That single property removes v1's whole class of phantom successes.

One-way imports

A module reaches another only through its public interface. Enforced by a build gate, not by good intentions.

Separable by config

Browser workers claim jobs over the database, so moving them to a second host is an environment change — never a code change.

The piece v1 did not have

Almost every v1 reliability failure traces to work that could run twice, could not resume, and could not honestly say what it had done.

Terminal states — each one distinct, each one true

succeeded The effect was confirmed — not attempted.	partial Some children completed, some did not.	failed Retryable attempts exhausted.
blocked A precondition is missing — no credentials, no capability, cost refused.	timed_out Exceeded its limit. Not the same as failed.	dead Dead-lettered with full context, requeueable.

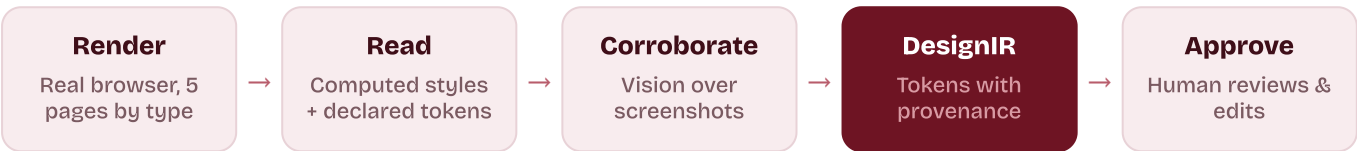
Eight guarantees

At-least-once	Lease with heartbeat. A worker that dies has its lease expire and the job is re-picked.
Effectively-once effects	Idempotency key → side-effect ledger. Recorded before attempt, checked before retry.
Atomic state + effect	State and its outbox event commit together. No window where a page is published and the ledger does not know.
Bounded retry	Backoff with jitter; retryable and terminal failures distinguished by class, not by string.
Dead letter	Full error chain, last provider response, one-click requeue.
Honest terminal state	Success requires confirmation of the effect.
Caps and fairness	Per-client concurrency and per-provider rate limits checked at lease time. One client's 50-page run cannot starve the rest.
Resumability	Parent/child ledger plus checkpointed AI graph state. A restart resumes at the first incomplete child.

Schedules ship disabled and are enabled one module at a time, only after that job kind survives the chaos suite. This is the deliberate inverse of v1, where the scheduler was off *because* the jobs were unsafe — and nobody could tell which.

Learn the client's design language, then speak it

The operator pastes the client's **previous website**. That site is the source of truth for the design language every new page must speak.



What DesignIR captures

Colour roles	Type scale	Spacing scale	Container & grid
Breakpoints	Radii & shadows	Buttons & CTAs	Form controls
Image treatment	Icon style	Nav & header	Footer structure
Section grammar	Section order	Motion	Coverage report

Provenance decides which source wins

declared	A CSS custom property the author wrote. It carries a name , which no clustering can recover. Highest rank.
derived	Clustered computed styles from the rendered page. Covers every role the declarations miss.
inferred	Extrapolated for a section the source site never had. Labelled as such to the reviewer.
defaulted	Nothing was found. Made visible, never hidden — v1's profiles silently collapsed to defaults.

Fidelity target: the same design system, a new composition.

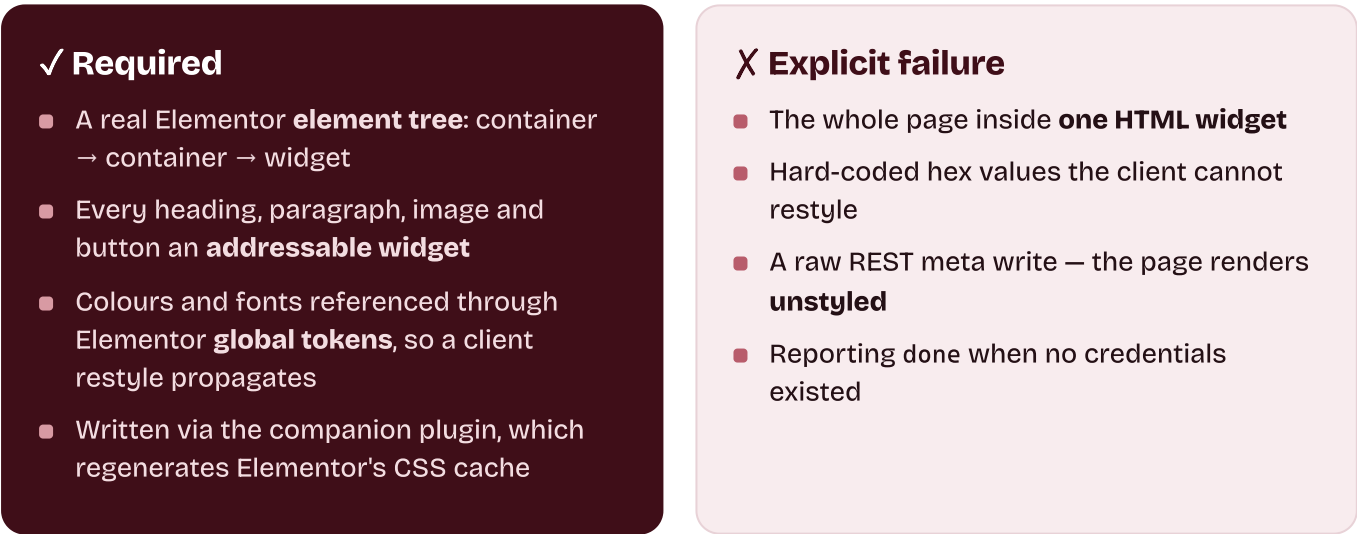
The test is **token conformance**, not a pixel diff — every colour, face, size, radius and spacing value on a generated page must exist in the approved DesignIR. Checked programmatically before publish. 100%, or it does not ship.

From keyword bank to an editable published page

The word bank is the SEO keyword bank with volumes. If a term is not in it, no page targets it and no tracker follows it.

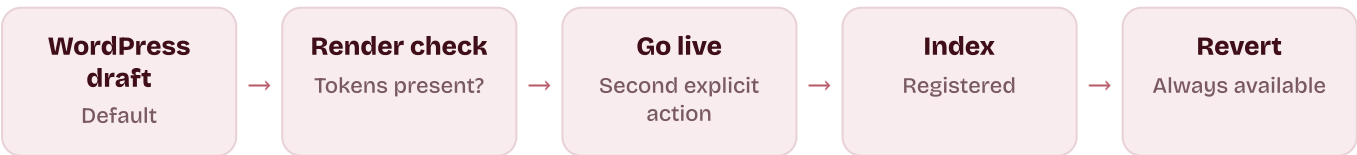


The publish requirement — and what fails it



Acceptance is physical: open the published page in Elementor, select any element, edit it, save. That is the test. v1 never built this, and it is the difference between a product and a demo.

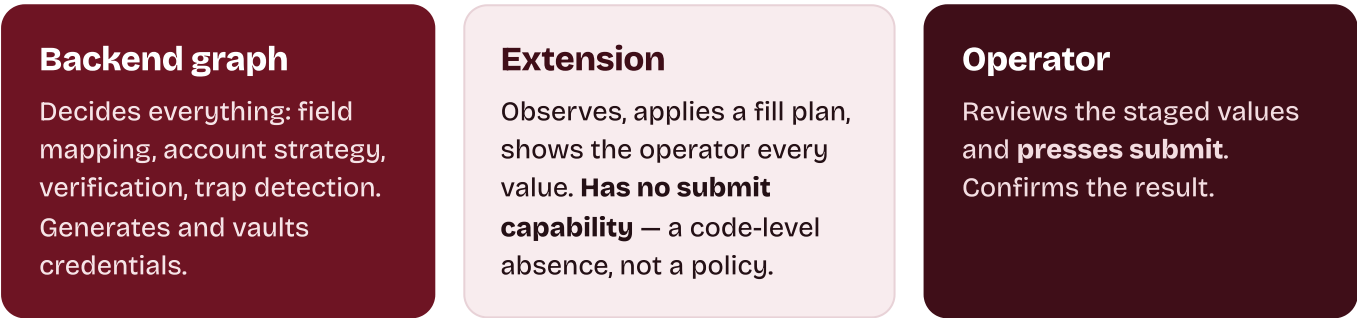
Publish flow



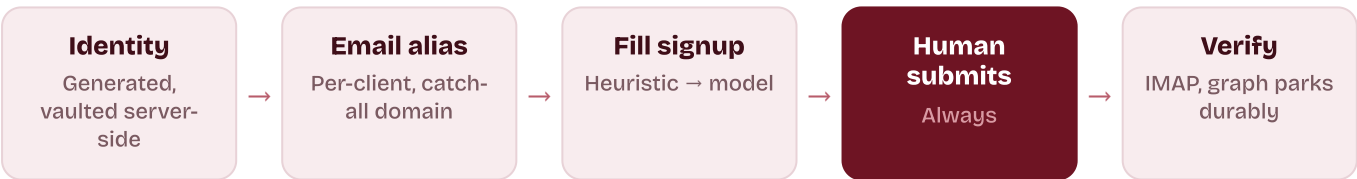
An agent that creates the account. A human who submits.

A browser extension as a thin actuator over a backend agent. All intelligence is server-side; the extension holds no credentials and no client data.

Division of labour



Account creation — the gap v1 never closed



Mappings cached on structure

The cache key is a fingerprint of **form shape** — field order, types, label shapes. Never the URL, never CSS selectors: directories rotate class names per load and reuse one form across many URLs.

A cache hit returns a plan in under 400 ms with **no model spend**. The fiftieth client on a directory costs almost nothing.

Honeypots: zero, not "low"

v1's filler put the client's website into a honeypot named `url` and **truthfully reported success**, while the directory silently discarded every submission.

Now: hidden, off-screen and decoy fields are classified **ignore**; the model reads "leave this empty" beside an input, which no heuristic can. Touching one is a **hard failure**. The eval bar is **0%**.

100

Live, NAP-verified citations — the release gate

25+

Directories with automated account creation

95%

Field-mapping accuracy on the fixture corpus

0%

Honeypot touches. Not a target — a gate

Thirty-two platforms. Official APIs only.

The reference surface is **SoMePoster**. We rebuild it multi-tenant and add the link-building spine a posting tool has no reason to have.

X	Facebook	LinkedIn	Instagram	Threads
Bluesky	Mastodon	Nostr	TikTok	YouTube
Pinterest	Blogger	WordPress.com	WP self-host	Ghost
DEV.to	Micro.blog	LiveJournal	Telegraph	GMB Profile
Webflow	Wix	Shopify	Mailchimp	Telegram
Discord	Slack	Pastebin	Google Drive	Diigo
Raindrop	n8n	Make.com	Medium — no API	+ registry

What we take from SoMePoster

- The connection grid and compose-once editor
- Video composer, AI studio, calendar, posts ledger
- **Parasite Poster** — keyword-targeted content on high-authority hosts

What we add

- Multi-tenancy with per-client identity and house-account caps
- Anchor bank with enforced distribution
- Internal linking, pacing, link liveness, account-health gating
- Approval and cost gates

A platform with no usable API is skipped, never automated. Medium is the worked example — its publishing API was withdrawn, so it is marked unsupported rather than scripted. Browser-automating a platform that forbids it risks the *client's* properties to save one integration.

The numbers a client keeps

This is where every integrity rule either pays off or fails publicly. Tracking output goes straight into client reports.

Three states, enforced by the database

ranked
Measured, and found at position N.

absent
Measured, and not present in the result set.

error
Never measured — rate limit, timeout, provider failure.

Collapsing **absent** and **error** renders a rate-limited afternoon as a client's service area collapsing — permanently, in an append-only table, in a report they keep. A CHECK constraint makes the blur impossible, and every ratio divides by **measured points only**.

Three sources, never mixed

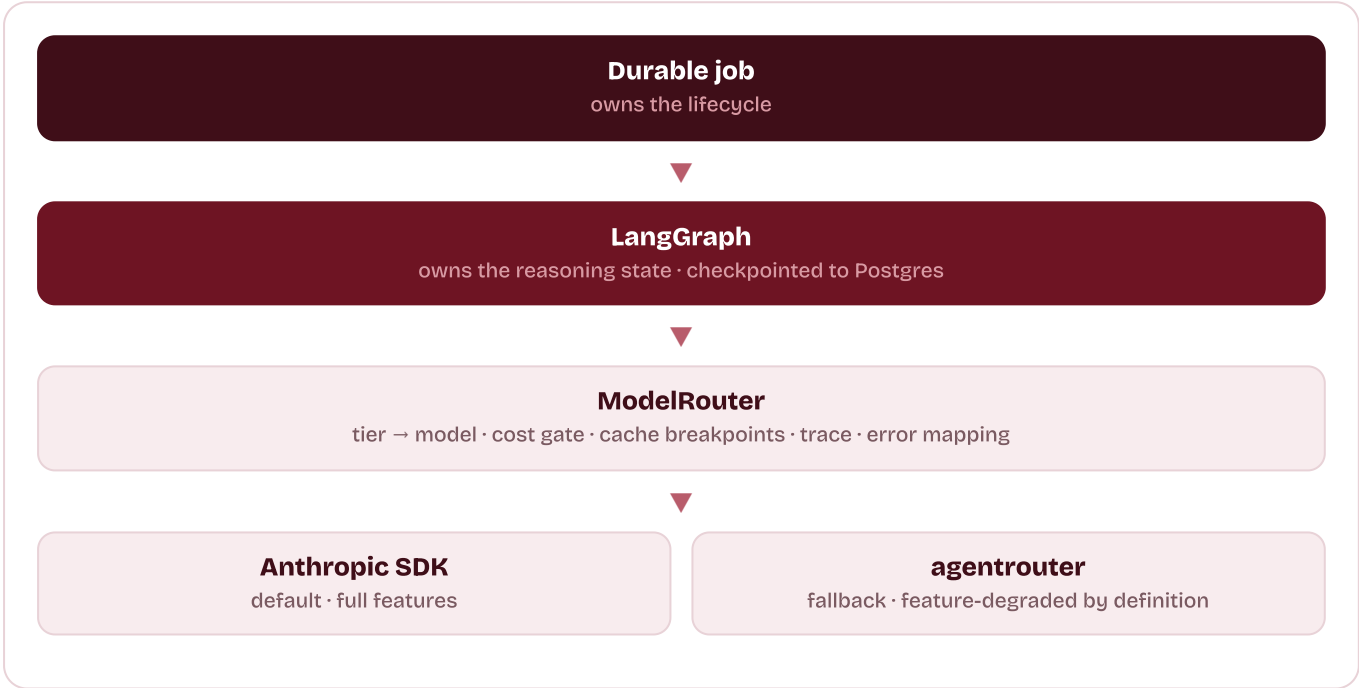
DataForSEO	Organic rank position, SERP features, competitors.	Never local pack or grid
serper.dev	Local pack presence and geo-grid points, queried by coordinate.	Never organic history
Search Console	First-party impressions, clicks, CTR, average position.	An average is not a rank

The v1 grid was fiction
It sent location="lat,lng" into a parameter that expects a place name, returning an unlocalised SERP for every point — a perfectly smooth, entirely fictional heat map. The fix is not cosmetic: the old data is worthless.

Publishing is not the deliverable
An **indexed** page is. Submissions that would exceed quota are **queued, never dropped**, and indexation is verified rather than assumed. unknown is a real state and is shown as one.

Orchestration by graph. Calls by our own router.

LangGraph earns its place for checkpointed, resumable state — the same property durable jobs provide. Both checkpoint to the same database.



Task tiers

reasoning · judge	Design reconciliation, page planning, audit narrative, QA judging, citation field mapping	Opus 5
drafting	Page bodies, Web 2.0 posts, GBP posts	Opus 5
structured	Schema extraction, classification, intent labelling	Sonnet 5
bulk	Alt text, meta variants, description variants, tags	Haiku 4.5

Caching is the cost model

A 50-page run reuses the same client context 50 times. Stable prefix first, volatile content after the last breakpoint, and cache-hit rate **charted per module** so a silent invalidator gets caught.

The judge is never the generator

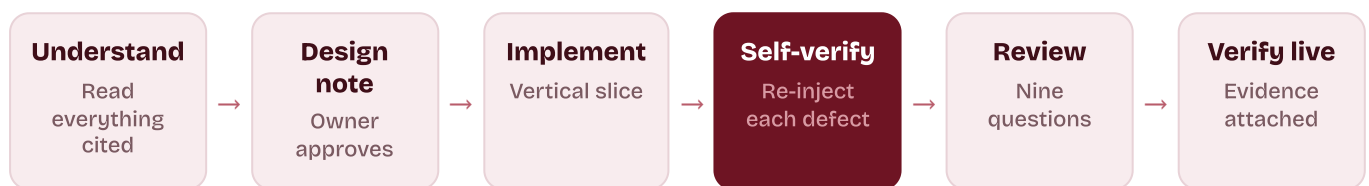
Evaluation uses a different prompt and a different tier configuration from the thing it grades. Six eval suites gate CI against fixed fixtures.

The agent cannot submit

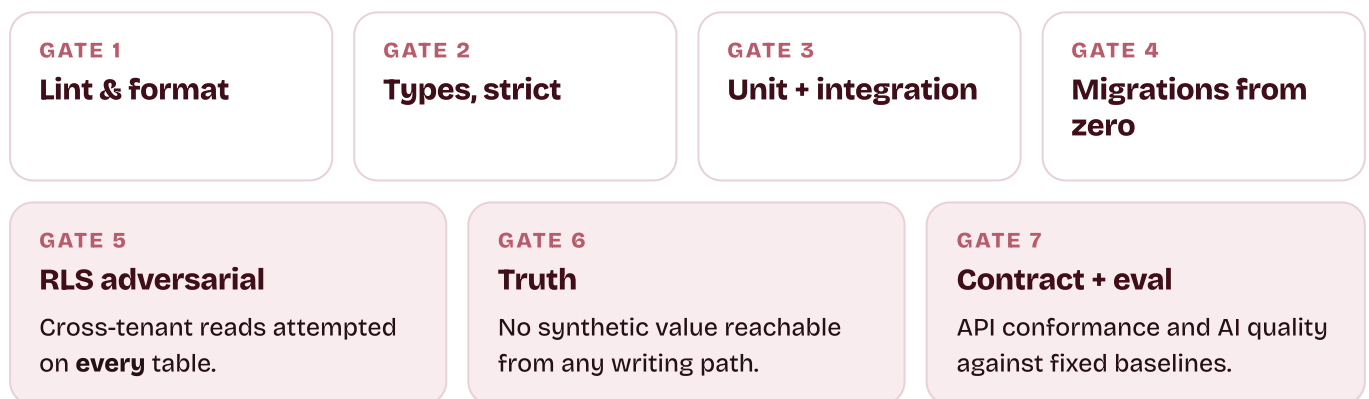
The citation graph has **no submit tool**. Scraped content enters prompts as clearly-delimited data, and nodes that read it have no tools at all.

Nothing lands that nobody can account for

There is one deployed environment. The process compensates: every migration is backward-compatible, every deploy is revertible by image tag, every module ships behind a flag.



The seven CI gates — none may be skipped or weakened



Definition of done

- Behaviour matches the requirement, **including its failure modes**
- Every failure mode has an honest terminal state
- Migration is forward-only and applies from zero
- New tenant tables have forced RLS and a policy
- Every paid call passes the cost gate
- Every test **fails when its defect is re-injected**
- Types clean, no unjustified escapes
- OpenAPI and the typed clients regenerated
- Traces, logs and metrics emitted
- Verified in the running system, with evidence**

Ten milestones. The order is load-bearing.

The two hardest things — emitting an editable Elementor tree and creating directory accounts at scale — are attempted early. If either is impossible, it is far cheaper to learn in week four than in month four.

- M0 Foundation** — repo, compose stack, RLS pattern, error taxonomy, auth, all seven gates failing loudly on seeded violations
- M1 Job engine & cost gate** — leases, idempotency, side-effect ledger, dead letters, vault, provenance, chaos suite v1
- M2 Vertical slice** — design analysis → DesignIR → **one page published as an editable Elementor tree.** *This milestone decides the product.*
- M3 Content at volume** — keyword research, topical map, 50-page fan-out, internal linking, QA calibration
- M4 Citations** — registry, extension, form intelligence, **account creation**, proof, liveness, cost accounting
- M5 Web 2.0 & social** — 32 platforms, ownership tiering, anchors, pacing, liveness, Parasite Poster
- M6 Measurement** — rank, geo-grid, Search Console, indexing, local & GBP
- M7 Audit** — six paid types plus the free lead magnet, findings contract, narrative constraints
- M8 Portal, reporting, tiers** — tasks, Command Center, capability truth table, client portal, document model, entitlements
- M9 Hardening & hand-over** — chaos across all job kinds, load at 100 clients, restore drill, runbooks, security checklist

If a milestone's exit criteria cannot be met, stop and re-plan. Do not proceed on a known-broken foundation with a note to fix it later — that is precisely how v1 arrived where it did.

Done is demonstrated, not declared

All eight must be true on production data, with evidence attached. A green test suite is not one of them.

S1 • CONTENT

50 pages, zero repair

Generated, Q'A'd, reviewed and published to a real site as editable Elementor trees in the client's design system, all passing token conformance.

S2 • CITATIONS

100 live listings

NAP-verified with proof artefacts, and a loaded cost per citation inside the agreed ceiling.

S3 • OFF-PAGE

One campaign, 15+ platforms

Per-client identities, images, internal links, correct SEO fields, official APIs only, no platform ban.

S4 • AUDIT

Six types, no degradation

Each paid audit type completes inside its cost ceiling with zero degraded sections on a fully-keyed environment.

S5 • ISOLATION

Adversarial reads all fail

Cross-tenant access attempted against every table in the schema, and refused by every one.

S6 • TRUTH

No unlabelled number

With every provider key removed, the system displays nothing that is not marked degraded — anywhere in the product.

S7 • RECOVERY

Eight job kinds survive a kill

Each resumes or dead-letters correctly under a mid-run worker kill, and none double-spends.

S8 • RESTORE

Backup restored, verified

Full restore into a clean host, passing a data-integrity check inside the documented recovery objective.

A requirement closed with "tests pass" and no runtime evidence is not closed.

Each closure records what was done, how it was verified in the running system, the evidence reference, and anything deliberately left undone.

Keep the knowledge. Replace the architecture.

v1 holds years of SEO domain knowledge worth porting. Its structure is the thing this rebuild exists to replace.

V1	V2
Sheets-as-store, then Postgres, then both	Postgres only. Sheets is an export format, never a store
Celery + Redis, empty schedule, no job contract	Postgres-backed durable jobs with idempotency, leases, dead letters
Direct model calls across ~40 services	One router + checkpointed graphs. Nothing else imports the SDK
Design capture ending in a styled HTML wrap	DesignIR → Elementor block tree with a conformance gate
Hand-maintained DOM specs per directory	Structural fingerprint + learned cache , plus account creation
Web 2.0 via shared house accounts and scrapers	32 platforms, official APIs, per-client identity where it matters
status="done" on a publish that published nothing	Honest terminal states , enforced by test
Synthetic providers reachable in production	Degrade-only , enforced by a CI gate

The three lessons, stated plainly

A test that cannot fail is worse than no test

It advertises coverage that does not exist, and it is why 4,900 tests did not catch a filler that failed silently.

Silent degradation is the expensive failure

Every v1 defect that reached a client was the system continuing confidently with data it did not have.

The job contract is the foundation

Not a detail. Work that can run twice, cannot resume, and cannot say what it did will produce every other failure eventually.

NEXT

Hand this pack to the build agent.

Sixteen documents, twelve module specifications, a numbered requirement register and a working agreement — enough context to start building in an empty repository without a single clarifying question.

READ FIRST

00 Executive scope
03 Architecture
12 Decision records
AGENTS.md

THEN BUILD

In the order given by
10 Backlog & roadmap.
The order is load-bearing.

PACK

docs/aio-v2/

REQUIREMENTS

130+ numbered

DECISIONS

21 recorded

PRECEDENCE

ADRs → pack → code